

Travis Ryan Dalton

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Education

Bachelor's degree in computer science, *Stratford University*

08/2007 – 05/2011

Baltimore

Professional Experience

Generative AI engineer, *Hitachi Digital Services*

01/2021 – 12/2024

San Francisco, California

- Designed and implemented generative AI solutions using **Stable Diffusion**, **GANs**, and **Flux**, developing advanced image generation, style transfer, and virtual try-on (VTON) systems.
- Fine-tuned large language models (LLMs) for **NLP tasks**, such as text generation, summarization, and question answering, leveraging frameworks like **Hugging Face Transformers** and **LangChain** for Retrieval-Augmented Generation (RAG).
- Built AI-driven applications using **LangChain** and **RAG** to integrate external knowledge bases with LLMs, enhancing contextual accuracy for enterprise-grade solutions.
- Developed and deployed **multi-agent systems (MAS)** and AI agents for autonomous decision-making, task management, and collaborative problem-solving in complex environments.
- Designed and optimized pipelines for fine-tuning LLMs and diffusion models, ensuring high performance and adaptability to domain-specific datasets.
- Built and maintained vector databases (e.g., **Pinecone**, **Weaviate**) and **PostgreSQL** backends for storing embeddings, improving semantic search and recommendation performance.
- Implemented and customized **ComfyUI nodes**, including **CatVTON** and **IBMVTON**, to enable interactive, user-driven workflows for generative design and virtual environments.
- Leveraged **AWS GenAI tools**, such as Bedrock, Claude AI, SageMaker, and AI custom endpoints, to build scalable and cost-efficient generative AI solutions for enterprise use cases.
- Deployed models using **CI/CD pipelines**, Docker, and Kubernetes, ensuring robust and scalable production-grade applications.
- Researched and applied state-of-the-art advancements in generative AI, staying at the forefront of industry trends and enhancing product innovation.

Senior machine learning engineer, *Apollo.io*

11/2015 – 12/2020

San Francisco, California

- Designed and deployed NLP models for SaaS platforms, including search ranking algorithms and content embeddings, using **PyTorch**, **TensorFlow**, and **Hugging Face Transformers**, improving search relevance and user experience by 35%.
- Fine-tuned transformer models (e.g., BERT, RoBERTa, DistilBERT) using the Hugging Face library for semantic search, personalized recommendations, and text classification tasks.

- Built and automated model evaluation frameworks to assess performance using metrics like precision, recall, F1-score, AUC-ROC, NDCG, and MRR for search ranking and recommendation systems.
- Leveraged **TensorFlow Serving**, **TorchServe**, and Hugging Face's model hub to deploy NLP models at scale, ensuring low-latency predictions in production environments.
- Conducted A/B testing and hyperparameter optimization to enhance model performance and user engagement.
- Built scalable machine learning pipelines on AWS (SageMaker, Lambda, S3, DynamoDB) to process and analyze large SaaS user datasets efficiently.
- Established automated monitoring and retraining pipelines for search ranking models, ensuring high relevance and adaptability to changing user behavior.

Data Scientist, Zillow

04/2012 – 10/2015
Seattle, Washington

- Built and maintained ETL pipelines to ingest and process large-scale real estate datasets, including property listings, sales transactions, and demographic data, ensuring data quality and reliability.
- Designed and deployed machine learning models for property price prediction and market trend analysis, integrating them into CI/CD pipelines for automated testing and deployment.
- Leveraged tools like Apache Airflow, SQL, and Python to streamline data workflows, reducing pipeline latency by 40%.
- Implemented CI/CD workflows using Jenkins and Docker to automate model deployment and ensure seamless integration with production environments.
- Developed dashboards and visualizations to provide actionable insights to stakeholders, driving data-driven decision-making in real estate investment and pricing strategies.

Python Developer, HTD Health

09/2011 – 04/2012
New York

- **Data Analysis:** Assisted in processing and analyzing healthcare datasets such as patient records and diagnostic data. Used Python libraries like NumPy for basic data manipulation and statistical analysis.
- **Automation & Scripting:** Developed Python scripts to automate data extraction, transformation, and loading (ETL) processes. Wrote scripts that interfaced with healthcare systems to streamline data input/output tasks.
- **Medical Data Processing:** Contributed to processing medical imaging data using early versions of OpenCV for basic image manipulation and processing tasks related to diagnostic data.
- **Database Management:** Assisted in storing healthcare data in relational databases like MySQL or PostgreSQL, using Python's built-in sqlite3 or MySQLdb for interfacing with databases.
- **Basic Machine Learning:** Worked on early-stage machine learning projects by implementing basic classification and regression algorithms using libraries like Scikit-learn. Assisted in building simple models to predict patient outcomes based on medical history.
- **Healthcare System Integration:** Helped integrate Python-based solutions with healthcare information systems, ensuring secure and accurate data handling according to industry standards like HIPAA.